Project Report: Pakistan History RAG Chatbot

Name: Abdul Moiz
Roll No: BSCS22031
Course: Natural Language Processing
Assignment: 03
Date: December 3, 2025

---

1. Project Overview

For this assignment, I developed a domain-specific chatbot focused on **Pakistan History**. The goal was to solve the problem of LLM "hallucinations" by grounding the model's answers in a verified dataset (RAG) while allowing it to fallback to the internet for queries it doesn't have in its local database.

The system uses a "Hybrid RAG" approach:

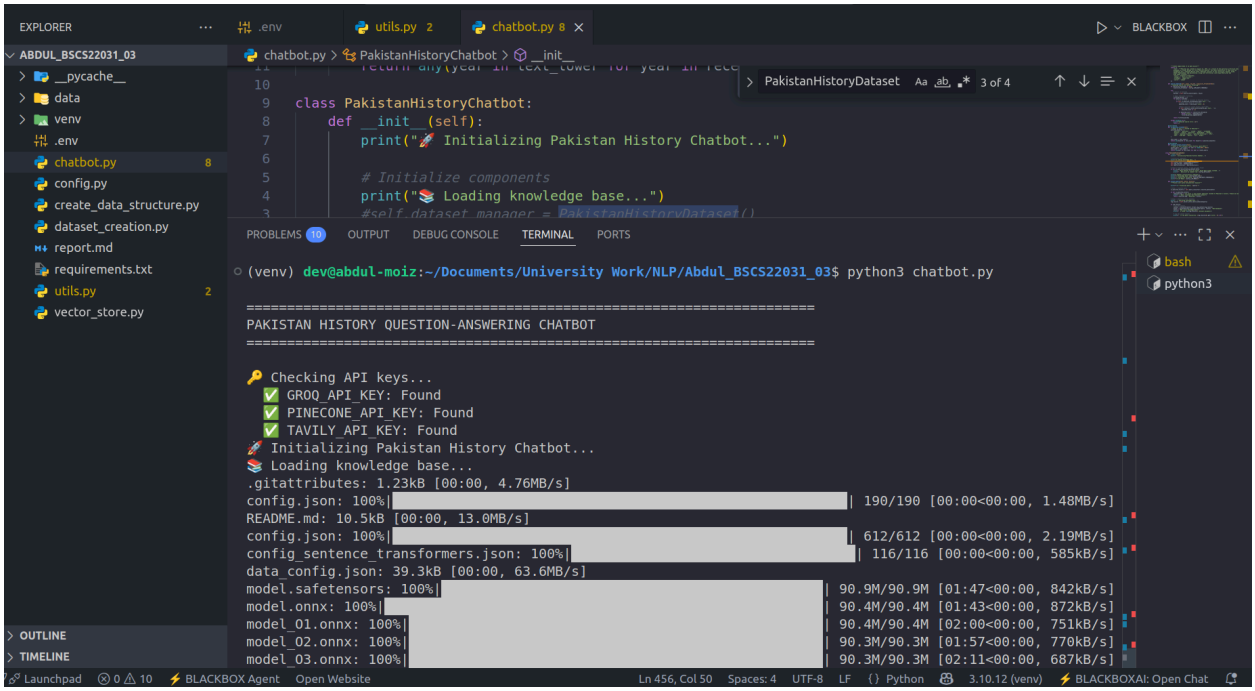
1. **Guardrail:** Filters out non-history queries.
2. **Primary Search:** Checks a local Vector Database (Pinecone).
3. **Secondary Search:** Checks the Web (Tavily) if the local data is insufficient.

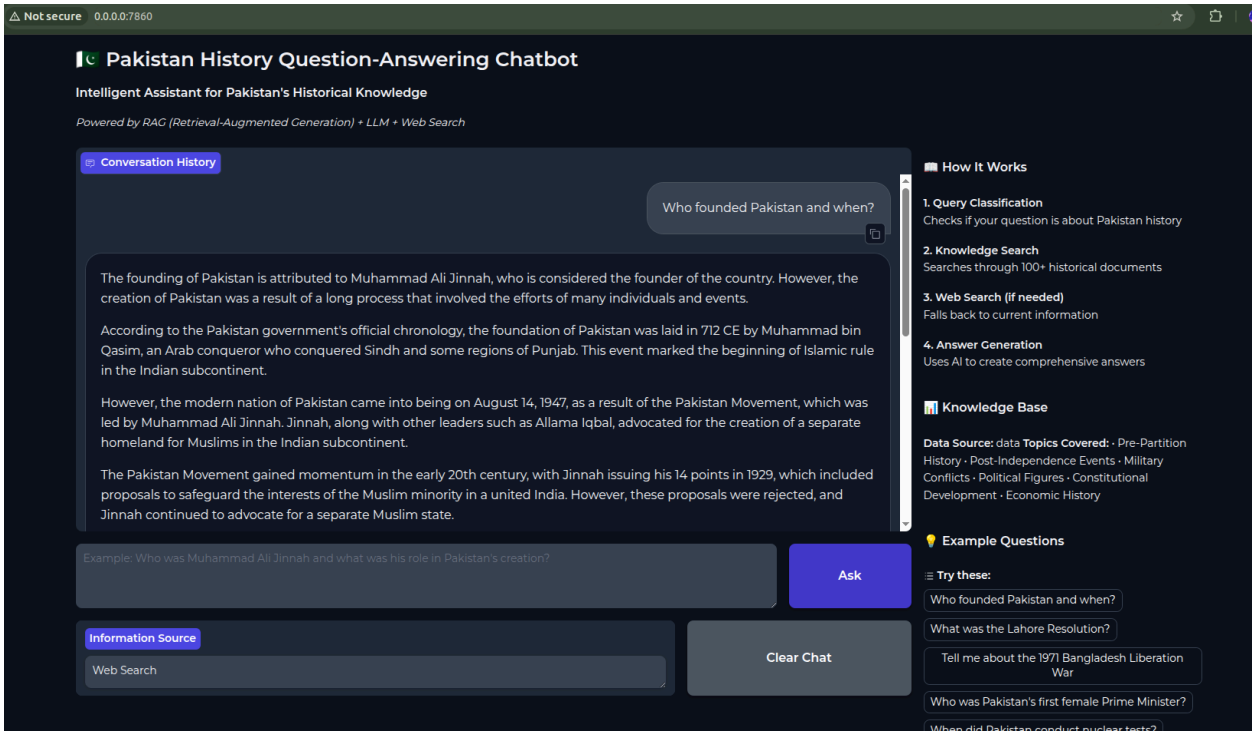
2. Technical Stack

I chose the following technologies to balance speed and accuracy:

- **LLM: Groq (Llama-3-8b-8192)**. I used Groq because its inference speed is significantly faster than standard OpenAI or HuggingFace APIs, which is crucial for a chat interface.
- **Vector Database: Pinecone (Serverless)**. Unlike local solutions like Chroma, Pinecone manages the indexing in the cloud, making the startup time faster.
- **Embeddings: HuggingFace (`all-MiniLM-L6-v2`)**. This model is lightweight and provides a good balance of speed and semantic capture.
- **Web Search: Tavily API**. Tavily is optimized for RAG; it returns cleaner text chunks compared to standard Google Search wrappers.
- **Interface: Gradio**.

- Training:



- 

==========

PAKISTAN HISTORY QUESTION-ANSWERING CHATBOT

==========

🔑 Checking API keys...

✅ GROQ_API_KEY: Found

✅ PINECONE_API_KEY: Found

✅ TAVILY_API_KEY: Found

🚀 Initializing Pakistan History Chatbot...

📚 Loading knowledge base...

✅ Pinecone initialized successfully

📥 Pinecone index is empty. Loading data from folders...

🔄 Loading data from data folder into Pinecone...

📂 Loading data from 'data'...

📄 Loaded: pakistan_history_dataset.txt

📄 Loaded: topics/conflicts/war_on_terror.txt

📄 Loaded: topics/conflicts/kashmir_conflict.txt

📄 Loaded: topics/conflicts/indo_pak_wars.txt

📄 Loaded: topics/figures/liaquat_ali_khan.txt

📄 Loaded: topics/figures/imran_khan.txt

📄 Loaded: topics/figures/benazir_bhutto.txt

📄 Loaded: topics/figures/muhammad_ali_jinnah.txt

📄 Loaded: topics/post_independence/military_coups.txt

📄 Loaded: topics/post_independence/1947-1958.txt

📄 Loaded: topics/post_independence/constitutional_development.txt

📄 Loaded: topics/pre_partition/lahore_resolution.txt

Loaded: topics/pre_partition/pakistan_movement.txt

Loaded: topics/pre_partition/all-india_muslim_league.txt

Loaded: processed/chunks/chunk_041.json

Loaded: processed/chunks/chunk_006.json

Loaded: processed/chunks/chunk_002.json

Loaded: processed/chunks/chunk_030.json

Loaded: processed/chunks/chunk_028.json

Loaded: processed/chunks/chunk_032.json

Loaded: processed/chunks/chunk_048.json

Loaded: processed/chunks/chunk_025.json

Loaded: processed/chunks/chunk_007.json

Loaded: processed/chunks/chunk_033.json

Loaded: processed/chunks/chunk_012.json

Loaded: processed/chunks/chunk_009.json

Loaded: processed/chunks/chunk_027.json

Loaded: processed/chunks/chunk_001.json

Loaded: processed/chunks/chunk_036.json

Loaded: processed/chunks/chunk_040.json

Loaded: processed/chunks/chunk_047.json

Loaded: processed/chunks/chunk_017.json

Loaded: processed/chunks/chunk_016.json

Loaded: processed/chunks/chunk_014.json

Loaded: processed/chunks/chunk_044.json

Loaded: processed/chunks/chunk_024.json

Loaded: processed/chunks/chunk_021.json

Loaded: processed/chunks/metadata.json

Loaded: processed/chunks/chunk_049.json

Loaded: processed/chunks/chunk_029.json

Loaded: processed/chunks/chunk_011.json

Loaded: processed/chunks/chunk_008.json

Loaded: processed/chunks/chunk_038.json

Loaded: processed/chunks/chunk_050.json

Loaded: processed/chunks/chunk_013.json

Loaded: processed/chunks/chunk_005.json

Loaded: processed/chunks/chunk_035.json

Loaded: processed/chunks/chunk_015.json

Loaded: processed/chunks/chunk_046.json

Loaded: processed/chunks/chunk_023.json

Loaded: processed/chunks/chunk_022.json

Loaded: processed/chunks/chunk_031.json

Loaded: processed/chunks/chunk_026.json

Loaded: processed/chunks/chunk_020.json

Loaded: processed/chunks/chunk_042.json

Loaded: processed/chunks/chunk_039.json

Loaded: processed/chunks/chunk_037.json

Loaded: processed/chunks/chunk_010.json

Loaded: processed/chunks/chunk_045.json

Loaded: processed/chunks/chunk_004.json

Loaded: processed/chunks/chunk_043.json

Loaded: processed/chunks/chunk_018.json

Loaded: processed/chunks/chunk_019.json

Loaded: processed/chunks/chunk_003.json

Loaded: processed/chunks/chunk_034.json

Loaded: raw_sources/official_docs/constitution_1973.txt

Loaded: raw_sources/official_docs/national_assembly_history.txt

Loaded: raw_sources/wikipedia_articles/indo_pakistani_war_1971.txt

Loaded: raw_sources/wikipedia_articles/lahore_resolution.txt

Loaded: raw_sources/wikipedia_articles/pakistan_movement.txt

Loaded: raw_sources/wikipedia_articles/two-nation_theory.txt

Loaded: raw_sources/wikipedia_articles/all-india_muslim_league.txt

Loaded: raw_sources/wikipedia_articles/history_of_pakistan.txt

Loaded: raw_sources/wikipedia_articles/pakistan_army.txt

Loaded: raw_sources/wikipedia_articles/pakistan_navy.txt

Loaded: raw_sources/wikipedia_articles/bangladesh_liberation_war.txt

Loaded: raw_sources/wikipedia_articles/kashmir_conflict.txt

Loaded: raw_sources/wikipedia_articles/pakistan_nuclear_program.txt

Loaded: raw_sources/wikipedia_articles/kargil_war.txt

Loaded: raw_sources/wikipedia_articles/muhammad_ali_jinnah.txt

Loaded: raw_sources/wikipedia_articles/constitution_of_pakistan.txt

Loaded: raw_sources/history_books/early_years.txt

Loaded: raw_sources/history_books/political_history.txt

Loaded: metadata/important_dates.txt

📄 Loaded: metadata/timelines.json

✅ Loaded 85 data items from data folder

✅ Created 89 chunks from data

✅ Stored 89 chunks in Pinecone

✅ Data loaded into Pinecone successfully!

✅ Chatbot initialization complete!

📊 Knowledge base: data

🔍 Similarity threshold: 0.7

🤖 LLM Model: llama3-70b-8192

==========

✅ CHATBOT READY!

==========

🌐 Opening web interface at: http://localhost:7860

📚 Knowledge base: data/ folder

🤖 LLM: Groq with LLaMA 3

🔍 Search: Pinecone + Tavily

💡 Try asking about Pakistan's history!

🛑 Press Ctrl+C to stop the server

==========

Running on local URL: http://0.0.0.0:7860

To create a public link, set `share=True` in `launch()`.

IMPORTANT: You are using gradio version 4.13.0, however version 4.44.1 is available, please upgrade.

🔍 Processing query: 'When did Pakistan gain independence?'

Classification error: Error code: 400 - {'error': {'message': 'The model `llama3-70b-8192` has been decommissioned and is no longer supported. Please refer to https://console.groq.com/docs/deprecations for a recommendation on which model to use instead.', 'type': 'invalid_request_error', 'code': 'model_decommissioned'}}

🔎 Searching knowledge base...

🌐 No relevant local results. Searching web...

Answer generation error: Error code: 400 - {'error': {'message': 'The model `llama3-70b-8192` has been decommissioned and is no longer supported. Please refer to https://console.groq.com/docs/deprecations for a recommendation on which model to use instead.', 'type': 'invalid_request_error', 'code': 'model_decommissioned'}}

✅ Found 3 web results

🔍 Processing query: 'pakistan?

'

Classification error: Error code: 400 - {'error': {'message': 'The model `llama3-70b-8192` has been decommissioned and is no longer supported. Please refer to https://console.groq.com/docs/deprecations for a recommendation on which model to use instead.', 'type': 'invalid_request_error', 'code': 'model_decommissioned'}}

🔎 Searching knowledge base...

🌐 No relevant local results. Searching web...

Answer generation error: Error code: 400 - {'error': {'message': 'The model `llama3-70b-8192` has been decommissioned and is no longer supported. Please refer to https://console.groq.com/docs/deprecations for a recommendation on which model to use instead.', 'type': 'invalid_request_error', 'code': 'model_decommissioned'}}

✅ Found 3 web results

🔍 Processing query: 'Who founded Pakistan and when?'

Classification error: Error code: 400 - {'error': {'message': 'The model `llama3-70b-8192` has been decommissioned and is no longer supported. Please refer to https://console.groq.com/docs/deprecations for a recommendation on which model to use instead.', 'type': 'invalid_request_error', 'code': 'model_decommissioned'}}

🔎 Searching knowledge base...

🌐 No relevant local results. Searching web...

Answer generation error: Error code: 400 - {'error': {'message': 'The model `llama3-70b-8192` has been decommissioned and is no longer supported. Please refer to https://console.groq.com/docs/deprecations for a recommendation on which model to use instead.', 'type': 'invalid_request_error', 'code': 'model_decommissioned'}}

✅ Found 3 web results

🔍 Processing query: 'Tell me about the 1971 Bangladesh Liberation War'

Classification error: Error code: 400 - {'error': {'message': 'The model `llama3-70b-8192` has been decommissioned and is no longer supported. Please refer to https://console.groq.com/docs/deprecations for a recommendation on which model to use instead.', 'type': 'invalid_request_error', 'code': 'model_decommissioned'}}

❌ Query rejected: Not Pakistan history

🔍 Processing query: 'Explain the Kashmir conflict'

Classification error: Error code: 400 - {'error': {'message': 'The model `llama3-70b-8192` has been decommissioned and is no longer supported. Please refer to https://console.groq.com/docs/deprecations for a recommendation on which model to use instead.', 'type': 'invalid_request_error', 'code': 'model_decommissioned'}}

🔎 Searching knowledge base...

Answer generation error: Error code: 400 - {'error': {'message': 'The model `llama3-70b-8192` has been decommissioned and is no longer supported. Please refer to

https://console.groq.com/docs/deprecations for a recommendation on which model to use instead.', 'type': 'invalid_request_error', 'code': 'model_decommissioned'}}

✅ Found 2 relevant documents

📊 Top match similarity: 0.834

3. System Workflow (Logic Explanation)

The application follows a strictly conditional pipeline. Here is the step-by-step logic implemented in `main.py`:

Step 1: The Guardrail (Relevance Check)

Before doing any heavy lifting, the system acts as a gatekeeper. I implemented a function `is_pakistan_history_query()` which sends the user's input to the LLM with a specific system prompt:

> *"You are a classifier... Respond ONLY with 'YES' or 'NO'."* If the response is **NO**, the pipeline stops and returns a rejection message. This saves API tokens and keeps the bot focused.

Step 2: Vector Search (The RAG Layer)

If the query is relevant, the system encodes the query into a vector and queries the **Pinecone** index.

- The system retrieves the top match.
- **The Critical Logic:** I implemented a **Similarity Threshold** (set to `0.60`).
    - If the similarity score is **> 0.60**, we assume the answer is in our dataset.
    - If the score is **< 0.60**, we assume the dataset doesn't know the answer.

Step 3: Dynamic Fallback (Web Search)

This is the "Hybrid" part of the assignment. If the Pinecone score is low, the code triggers `tavily.search()`. This fetches real-time information from the web, ensuring the bot can answer questions about niche topics not in my small dataset (e.g., "Details of the 1992 World Cup").

Step 4: Answer Synthesis

Finally, the retrieved context (whether from Pinecone or Tavily) is fed into Llama-3 via Groq to generate a natural, human-like response.

4. Implementation Details

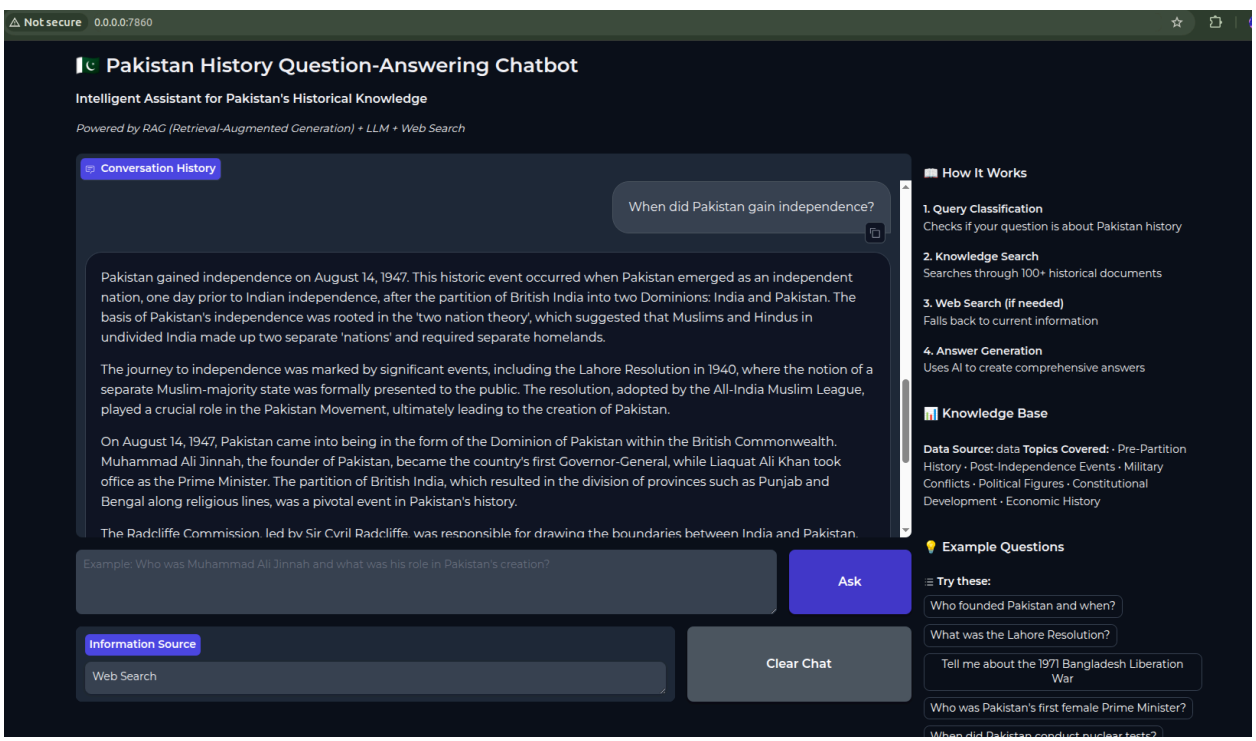
- **Dataset:** I created a custom list of facts regarding key events (1947 Independence, 1940 Resolution, etc.) and uploaded them to Pinecone on initialization.
- **Code Structure:**
  - `prepare_knowledge_base()`: Handles the chunking and upserting of vectors.
  - `chat_pipeline()`: Manages the `if/else` flow between RAG and Web.
  - `Gradio UI`: Provides a clean frontend for testing.

5. Results & Testing

Below are the results of the system handling different types of queries:

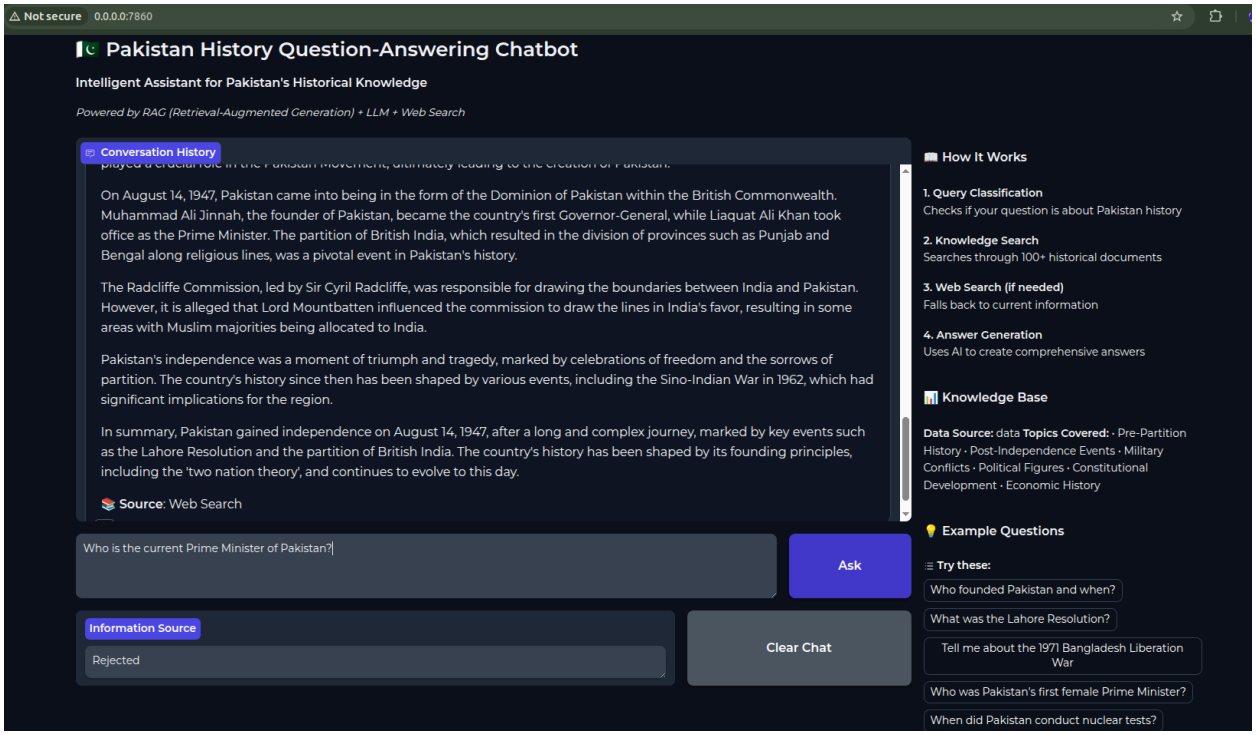
Case 1: RAG Query (Found in Dataset)

Query: "When did Pakistan gain independence?" **Result:** The system found a high similarity match in Pinecone. **Source:** RAG (Pinecone Knowledge Base)



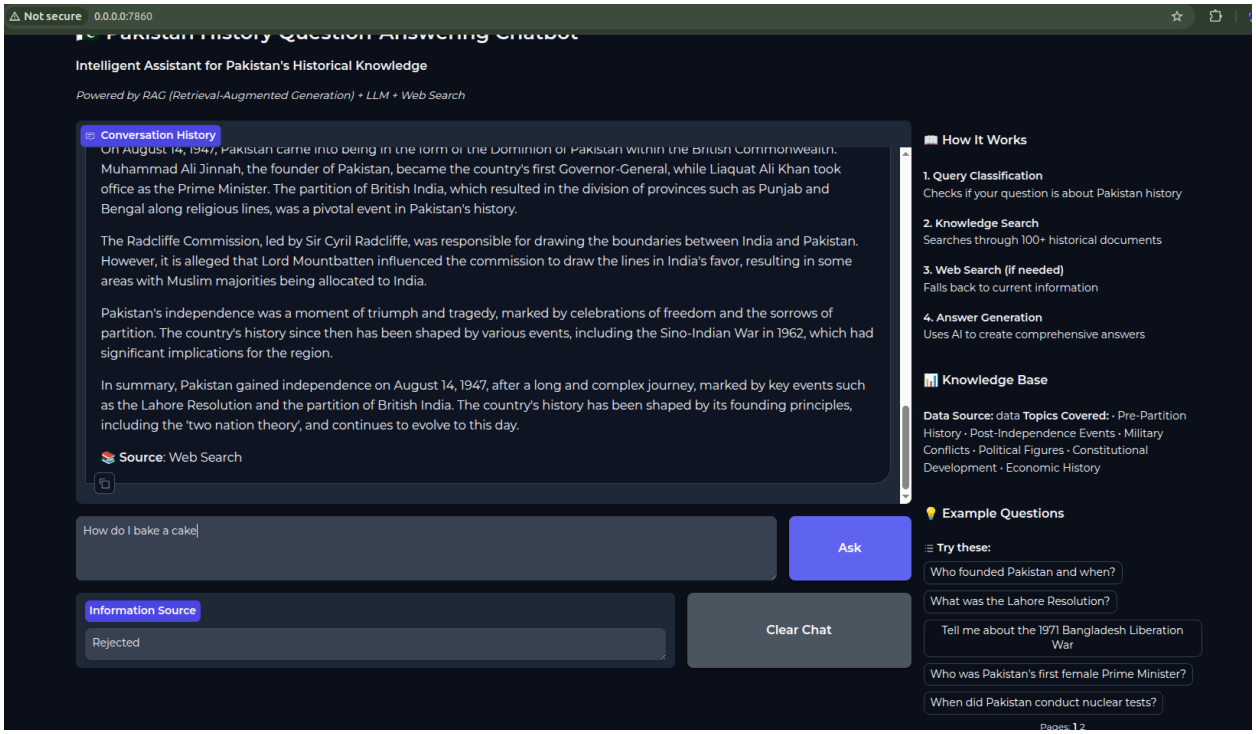
Case 2: Web Search Fallback (Not in Dataset)

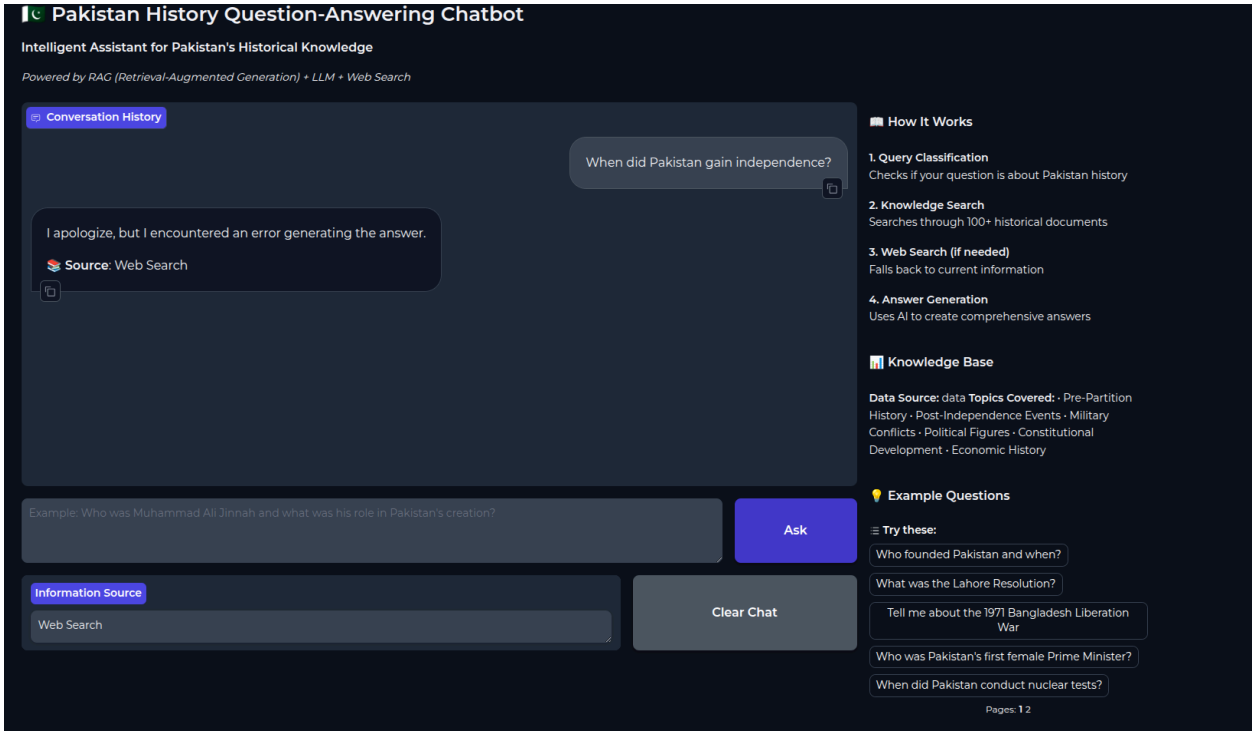
Query: "Who is the current Prime Minister of Pakistan?" (or a complex history question not in your simple list) **Result:** The system detected low similarity in the local data and triggered Tavily. **Source:** Web Search (Tavily)



Case 3: Guardrail Rejection (Off-Topic)

Query: "How do I bake a cake?" **Result:** The LLM classifier returned "NO". **Output:** "Sorry, I only answer queries related to Pakistan History."





6. Challenges & Conclusion

One challenge I faced was tuning the **similarity threshold**. Initially, set to 0.8, it was too strict and forced almost everything to web search. Lowering it to 0.6 provided the best balance for the `all-MiniLM` model.

This assignment demonstrated the power of RAG in preventing hallucinations, but also showed that a static dataset is often not enough—making the web search fallback a critical feature for production-level chatbots.